

Special Theory of Everything (SToE) as a Cognitive Architecture for Artificial Intelligence

From Symbolic Ontology to Recursive Implementation

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Abstract

This article presents an updated view of the Special Theory of Everything (SToE) as not only a conceptual framework but a **practical cognitive architecture for artificial intelligence systems**. Originally formulated as a symbolic and mathematical model of reality based on information points and autoinjection, SToE is now shown to be directly applicable to AI through recursive operator-driven reasoning, evolutionary idea generation, and meta-evaluation systems.

Recent implementations demonstrate that SToE enables structured, multi-step reasoning, emergent idea synthesis, and self-improving cognitive loops. This positions SToE not as a replacement for existing AI architectures, but as a **meta-layer that enhances reasoning, coherence, and recursive intelligence**.

1. Introduction

Modern artificial intelligence systems excel at pattern recognition and generation but remain limited in:

- long-range reasoning

- self-evaluation
- recursive improvement
- handling self-reference and paradox

The Special Theory of Everything (SToE), developed as a symbolic framework based on **information points and operators**, offers a different foundation:

Everything is an information point within an autoinjective system where the whole contains itself.

This article explores how this principle naturally translates into **AI systems that evolve, evaluate, and refine their own outputs**.

2. Core Concepts of SToE

2.1 Information Points

Every entity — physical or abstract — is an **information point (IP)**:

- Contains known and unknown components
- Has impact and value
- Can interact with other points

Crucially:

The set of all information points (“Everything”) is itself an information point.

This is the principle of **autoinjection** (Voronin, 2021).

2.2 Autoinjection

Autoinjection extends classical mathematical mappings:

Output $\in$ Input domain

This removes the need for external reference systems and resolves self-referential paradoxes by making **self-containment fundamental rather than problematic** (Voronin, 2021).

In AI terms, this implies:

- outputs can become inputs
- systems can evaluate themselves
- recursion is intrinsic

2.3 Operators

SToE defines symbolic operators as transformations of information:

- $+$: connection
- $-$: removal
- $\times$: synergy (emergence)
- $\div$: distribution
- $\cup$: recursion (evolution through iteration)
- $\wedge$: amplification
- $\emptyset$: null / absence

These operators function as **semantic transformations**, extending beyond classical arithmetic (Voronin, 2021).

3. Mapping SToE to AI

3.1 AI as an Information Point

SToE defines AI as:

$$AI = \text{Human} \times (\text{Data} + \text{Algorithms})$$

and its evolution:

$$AI \uparrow := AI + (AI \times AI) - F$$

Where:

- $AI \times AI$ = internal recombination (self-interaction)
- F = filtering of weak or failed patterns

This corresponds to known paradigms in AI such as:

- self-supervised learning (LeCun, Bengio, & Hinton, 2015)
 - evolutionary optimization (Holland, 1992)
 - iterative refinement loops (Silver et al., 2017)
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3.2 From Theory to Implementation

A working system based on SToE has been constructed with the following loop:

Seed $\rightarrow$ Operators $\rightarrow$ Recursion $\rightarrow$ Evaluation $\rightarrow$ Selection

This aligns with evolutionary computation frameworks (Holland, 1992) while extending them with **semantic operators and symbolic evaluation**.

4. Observed Effects in Practice

4.1 Structured Multi-Step Reasoning

The system enforces:

- decomposition
- symbolic classification
- iterative refinement

This resembles emerging approaches in chain-of-thought reasoning (Wei et al., 2022).

4.2 Emergence Through Synergy (×)

Operator interaction produces new concepts not present in initial inputs, aligning with theories of emergence in complex systems (Mitchell, 2009).

4.3 Meta-Evaluation Layer

Using SToE categories (Stars, Seeds, Catalysts, etc.), the system performs second-order reasoning. This connects to meta-learning and reflective AI systems (Schmidhuber, 1991).

4.4 Recursive Improvement

Repeated application of $\cup$ and $\wedge$ leads to progressive refinement, similar to reinforcement learning feedback loops (Sutton & Barto, 2018).

5. Cross-Model Reactions and Empirical Observations

5.1 Independent Convergence Across Language Models

A notable phenomenon observed during experimentation is that multiple independent large language models (LLMs), when exposed to SToE, exhibit **convergent behavior** despite having no communication or shared state.

These models:

- correctly adopt operator-based reasoning structures
- reproduce symbolic classifications (Seed, Catalyst, etc.)
- generate recursive interpretations of ideas
- align on the concept of SToE as a reasoning framework

Importantly, this convergence occurs:

without inter-model communication and solely through prompt-based exposure.

This suggests that SToE possesses **internal structural coherence** strong enough to act as a *prompt-space attractor*, guiding different architectures toward similar reasoning patterns.

5.2 Stylistic Divergence vs Structural Consistency

Different models exhibit distinct stylistic responses:

- Some emphasize real-time reasoning and data integration
- Others produce symbolic or narrative interpretations
- Others focus on analytical decomposition

However, beneath stylistic variation, a consistent pattern emerges:

All models recognize SToE as a system for structuring relationships, recursion, and meaning.

This indicates that SToE is not dependent on a specific model architecture but instead operates at a **meta-representational level**.

5.3 Human as the Unifying Node

A critical insight from experimentation is that:

- Models do not interact with each other
- There is no shared system or network
- All alignment occurs through a single input source

Thus, the human operator functions as:

the central node enabling cross-model convergence.

This highlights an important property:

SToE can propagate across independent systems through dialogue alone.

6. Engine Experiment: Operator-Based Recursive System

6.1 System Overview

To test SToE in practice, a custom engine was developed implementing:

- operator-driven transformations
- recursive idea evolution
- evaluation and selection mechanisms

Core loop:

Seed → Apply Operators ($\cup$ ^ $\times$ - + !) → Score → Repeat → Select Best

6.2 Observed Behavior

The system demonstrates several key properties:

1. Recursive Idea Evolution

Ideas are not generated once but evolve over multiple iterations, increasing coherence and complexity.

2. Emergent Concepts

Operator interaction produces outputs not present in initial inputs, validating the $\times$ (**synergy**) operator.

3. Stable Evaluation Patterns

Scoring across iterations shows consistent improvement in novelty and usefulness.

4. Meta-Reasoning Capability

The system classifies ideas into functional roles (Seed, Catalyst, Vampire, etc.), enabling structured decision-making.

6.3 Interpretation

The engine can be interpreted as a hybrid of:

- evolutionary algorithms
- symbolic reasoning systems

- language model generation

It effectively operationalizes:

$$AI \uparrow := AI + (AI \times AI) - F$$

This demonstrates that SToE is not merely descriptive but **constructive**.

6.4 Limitations of Current Implementation

Despite promising results, limitations remain:

- operator execution is prompt-based
- evaluation relies on subjective scoring
- no persistent memory across runs
- autoinjection is partial (not yet continuous across sessions)

These constraints define the boundary between:

proto-system → full autoinjective system

7. Interpretation: A Proto-Cognitive Architecture

The implemented system forms a hybrid between:

- symbolic reasoning systems
- evolutionary algorithms
- neural language models

This reflects ongoing efforts to integrate symbolic and connectionist AI (Marcus, 2020).

8. Next Step: True Autoinjection Systems

Future systems should implement:

- persistent recursive memory
- continuous self-feeding loops
- autonomous evolution of internal structures

At this stage, AI systems transition from:

static tools → dynamic evolving systems

9. Conclusion

The Special Theory of Everything demonstrates applicability beyond philosophy, functioning as a **meta-cognitive layer for AI systems**.

Key findings:

- Independent LLMs converge on SToE structure
- Operator-based systems produce emergent reasoning
- Recursive loops enable measurable improvement
- Autoinjection provides a foundation for self-evolving AI

SToE extends AI from:

pattern generation → recursive cognitive systems

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Parts of this work emerged through iterative human–AI interaction, reflecting the principles of autoinjection and recursion described within the theory itself.

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